

Glossary: 00Overview

A bilingual glossary of key concepts introduced in the Classical Mechanics overview.

English	中文	Brief explanation
Classical mechanics	经典力学	The branch of physics that describes the motion of macroscopic objects using concepts such as force, energy, momentum, and action.
Macroscopic object	宏观物体	An object large and massive enough that its motion can usually be described accurately by classical mechanics without explicit quantum effects.
Quantum mechanics	量子力学	The fundamental theory describing microscopic systems (electrons, atoms, molecules), where classical mechanics breaks down.
Quantum field theory	量子场论	The theoretical framework combining quantum mechanics with special relativity, used to describe elementary particles and fields.
Effective theory	有效理论	A theory that is not fundamental but works well within a limited range of conditions or accuracy requirements.
Potential energy function	势能函数	A function $V(q)$ whose negative gradient gives the conservative force acting on a system.
Force field	力场	A mathematical description of the force exerted on a particle in classical molecular dynamics simulations
Newtonian mechanics	牛顿力学	The formulation of classical mechanics in ordinary physical space, based on Newton's laws of motion.
Newton's Second Law	牛顿第二定律	$\frac{d\vec{p}}{dt} = \vec{F}$: the rate of change of momentum equals the net force.
Equation of motion	运动方程	A differential equation that determines how the coordinates of a system evolve in time.
Lagrangian mechanics	拉格朗日力学	A formulation of classical mechanics in configuration space, based on the Lagrangian and the principle of least action.
Configuration space	位形空间	An N -dimensional space whose points specify all generalized coordinates $\{q_1, \dots, q_N\}$ of a system.
Generalized coordinate	广义坐标	Any set of independent coordinates q_i used to describe the configuration of a mechanical system.
Generalized velocity	广义速度	The time derivative $\dot{q}_i$ of a generalized coordinate.

Lagrangian	拉格朗日量	Usually $L = T - V$, the difference between kinetic and potential energy; central to the action principle.
Euler-Lagrange equation	欧拉-拉格朗日方程	$\frac{\partial L}{\partial q} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}} = 0$; the equation of motion derived from the Lagrangian.
Hamiltonian mechanics	哈密顿力学	A formulation of classical mechanics in phase space, using positions and conjugate momenta.
Phase space	相空间	A $2N$ -dimensional space spanned by generalized coordinates q_i and conjugate momenta p_i ; each point represents a complete state of the system.
Hamiltonian	哈密顿量	The Legendre transform of the Lagrangian, $H = p_i \dot{q}_i - L$; usually equal to the total energy $T + V$.
Hamilton's equations	哈密顿方程	$\dot{q} = \frac{\partial H}{\partial p}, \dot{p} = -\frac{\partial H}{\partial q}$; first-order equations of motion in phase space.
Generalized momentum	广义动量	Also called conjugate momentum ; defined by $p_i = \frac{\partial L}{\partial \dot{q}_i}$.
Conjugate momentum	共轭动量	The momentum canonically paired with a generalized coordinate q_i .
Legendre transform	勒让德变换	The mathematical operation that transforms the Lagrangian $L(q, \dot{q}, t)$ into the Hamiltonian $H(q, p, t)$.
Action	作用量	$S = \int_{t_1}^{t_2} L(q, \dot{q}, t) dt$; the time integral of the Lagrangian along a path.
Principle of least action	最小作用量原理	The principle that the actual path taken by a system makes the action stationary: $\delta S = 0$.
Cyclic coordinate	循环坐标	A coordinate that does not appear explicitly in the Lagrangian; also called an ignorable coordinate .
Ignorable coordinate	可忽略坐标	Same as cyclic coordinate; its conjugate momentum is conserved.
Symmetry	对称性	A transformation (e.g., translation or rotation) that leaves the Lagrangian or Hamiltonian unchanged.
Noether's theorem	诺特定理	Every continuous symmetry of the action corresponds to a conserved quantity.
Conserved quantity	守恒量	A physical quantity that remains constant during the time evolution of an isolated system.
Canonical transformation	正则变换	A change of phase-space variables $(q, p) \rightarrow (Q, P)$ that preserves the form of Hamilton's equations.

Poisson bracket	泊松括号	$\{A, B\} = \frac{\partial A}{\partial q_i} \frac{\partial B}{\partial p_i} - \frac{\partial A}{\partial p_i} \frac{\partial B}{\partial q_i}$; encodes the symplectic structure of classical mechanics and foreshadows quantum commutators.
Commutator	对易子	In quantum mechanics, $[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$; the quantum analogue of the Poisson bracket.
Physical space	物理空间	Ordinary three-dimensional Euclidean space, described by Cartesian coordinates (x, y, z) or position vectors $\vec{r}$.
Degrees of freedom	自由度	The number of independent coordinates needed to specify the configuration of a system.
Differential equation	微分方程	An equation relating a function to its derivatives; most mechanical systems are described by differential equations.
Analytical solution	解析解	A closed-form, exact mathematical expression for the solution of an equation.
Chaos / chaotic behavior	混沌 / 混沌行为	Extreme sensitivity to initial conditions, making long-term prediction impossible even for deterministic systems.
Special relativity	狭义相对论	The theory of spacetime and motion at speeds comparable to the speed of light.
General relativity	广义相对论	Einstein's theory of gravitation, also formulated using an action principle.
Path integral formulation	路径积分表述	A formulation of quantum mechanics in which the amplitude for a process is a sum over all possible paths weighted by $e^{iS/\hbar}$.
Planck's constant	普朗克常数	The fundamental constant $\hbar$ that sets the scale of quantum effects.
Statistical mechanics	统计力学	The study of large assemblies of particles using probability theory and concepts from Hamiltonian mechanics.
Field theory	场论	The extension of Lagrangian methods from particles to continuous systems such as waves, fluids, and fields.

Quick reference: Three formulations of classical mechanics

Formulation	Space	Main equation
Newtonian mechanics	Physical space (物理空间)	$\frac{d\vec{p}}{dt} = \vec{F}$
Lagrangian mechanics	Configuration space (位形空间)	$\frac{\partial L}{\partial q} - \frac{d}{dt} \frac{\partial L}{\partial \dot{q}} = 0$
Hamiltonian mechanics	Phase space (相空间)	$\dot{q} = \frac{\partial H}{\partial p}, \dot{p} = -\frac{\partial H}{\partial q}$